

General equations

Emitted field equations (emitted field is converted to Rabi frequency, hence it has units of [frequency]) :

$$\left[\frac{\partial}{\partial z} - \frac{i}{2k_0} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \Omega_\sigma^{(+)}(\vec{r}, t) = -\frac{\kappa_{\Omega_\sigma}}{2} \Omega_\sigma^{(+)} + i \frac{3}{8\pi} \lambda^2 \Gamma_{sp} \left(\left[\sum_{i,j} n H_{ij} T_{ji\sigma} \frac{\rho_{ij}(\vec{r}, t) + \rho_{ji}^*(\vec{r}, t)}{2} \right] + h_\sigma(\vec{r}, t) \xi_\sigma^{(+)}(\vec{r}, t) \right) \quad (1)$$

$$\left[\frac{\partial}{\partial z} + \frac{i}{2k_0} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \Omega_\sigma^{(-)}(\vec{r}, t) = -\frac{\kappa_{\Omega_\sigma}}{2} \Omega_\sigma^{(-)} - i \frac{3}{8\pi} \lambda^2 \Gamma_{sp} \left(\left[\sum_{i,j} n H_{ji} T_{ji\sigma} \frac{\rho_{ij}(\vec{r}, t) + \rho_{ji}^*(\vec{r}, t)}{2} \right] + h_\sigma(\vec{r}, t) \xi_\sigma^{(-)}(\vec{r}, t) \right) \quad (2)$$

Pump field equation with diffraction:

$$\left[\frac{\partial}{\partial z} - \frac{i}{2k_P} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \mathcal{P} = -\frac{\kappa_P}{2} \mathcal{P}, \quad (3)$$

Atomic equations:

$$\begin{aligned} \frac{\partial}{\partial t} \rho_{ij}(\vec{r}, t) = & -i \left(\frac{E_i - E_j}{\hbar} - \text{sign}_{ij} \omega_0 \right) \rho_{ij}(\vec{r}, t) \\ & - \left(\sum_s \frac{\gamma_{si} + \gamma_{sj}}{2} \right) \rho_{ij}(\vec{r}, t) + \delta_{ij} \sum_s \gamma_{is} \rho_{ss}(\vec{r}, t) \\ & + i \sum_{\sigma, s} \rho_{sj}(\vec{r}, t) T_{is\sigma} [H_{is} \Omega_\sigma^{(+)}(\vec{r}, t) + H_{si} \Omega_\sigma^{(-)}(\vec{r}, t)] \\ & - i \sum_{\sigma, s} \rho_{is}(\vec{r}, t) T_{sj\sigma} [H_{sj} \Omega_\sigma^{(+)}(\vec{r}, t) + H_{js} \Omega_\sigma^{(-)}(\vec{r}, t)] \\ & + \sum_{\sigma, s} \tilde{h}_\sigma(\vec{r}, t) [H_{si} \rho_{sj}(\vec{r}, t) T_{is\sigma} \xi_\sigma^{(+)*}(\vec{r}, t) + H_{sj} \rho_{is}(\vec{r}, t) T_{sj\sigma} \xi_\sigma^{(-)*}(\vec{r}, t)] \end{aligned} \quad (4)$$

Choice of noise multiplicative factors:

$$\begin{aligned} h_\sigma(\vec{r}, t) &= \sqrt{\frac{T_{i'j'\sigma}^{(-)} \rho_{j'k'}(\vec{r}, t) T_{k'i'\sigma}^{(+)} n}{\gamma_{decoh.} \Delta x \Delta y \Delta z \Delta t}}, \\ \tilde{h}_\sigma(\vec{r}, t) &= \frac{1}{h_\sigma(\vec{r}, t) \Delta x \Delta y \Delta z \Delta t} = \sqrt{\frac{\gamma_{decoh.}}{T_{i'j'\sigma}^{(-)} \rho_{j'k'}(\vec{r}, t) T_{k'i'\sigma}^{(+)} n \Delta x \Delta y \Delta z \Delta t}}. \end{aligned} \quad (5)$$

Noise correlation properties

$$\begin{aligned}\langle \xi_{\sigma}^{(\pm)}(x_i, y_j, z_k, t_l) \xi_{\sigma'}^{(\pm)*}(x_{i'}, y_{j'}, z_{k'}, t_{l'}) \rangle &= \delta_{\sigma\sigma'} \delta_{ii'} \delta_{jj'} \delta_{kk'} \delta_{ll'} \\ \langle \xi_{\sigma}^{(\pm)}(\vec{r}, t) \xi_{\sigma'}^{(\pm)*}(\vec{r}', t') \rangle &= \langle \xi_{\sigma}^{(\pm)*}(\vec{r}, t) \xi_{\sigma'}^{(\pm)}(\vec{r}', t') \rangle = \langle \xi_{\sigma}^{(\mp)}(\vec{r}, t) \xi_{\sigma'}^{(\pm)*}(\vec{r}', t') \rangle = 0\end{aligned}\quad (6)$$

here

$$\text{sign}_{ij} = \begin{cases} +1 & \text{if } E_i > E_j \\ 0 & \text{if } i = j \\ -1 & \text{if } E_i < E_j \end{cases}, \quad H_{ij} = \begin{cases} +1 & \text{if } E_i > E_j \\ 0 & \text{if } E_i \leq E_j \end{cases}, \quad (7)$$

polarizations are denoted by index σ , the tensor $T_{ij\sigma}$ is dimensionless tensor representing the ratio between dipole matrix elements between transitions (corresponding to levels i, j) and couplings to field with polarization σ .

In terms of the Rabi frequency, the emitted electric field is expressed as:

$$\mathcal{E}_{\sigma}^{(\pm)} = \Omega_{\sigma}^{(\pm)} \sqrt{\frac{\hbar k_0^3}{3\pi\epsilon_0\Gamma_{sp}}}, \quad (8)$$

the corresponding photon flux is

$$J_{\Omega_{\sigma}} = \frac{dN_{ph,\sigma}}{dtdS} = \frac{2\epsilon_0 c \mathcal{E}_{\sigma}^{(+)} \mathcal{E}_{\sigma}^{(-)}}{\hbar\omega_0} = \frac{\Omega_{\sigma}^{(+)} \Omega_{\sigma}^{(-)}}{\frac{3}{8\pi} \lambda^2 \Gamma_{sp}}, \quad (9)$$

here the spontaneous emission rate is assumed to be due to all transitions (e.g., both Cu-K $_{\alpha 1}$ and Cu-K $_{\alpha 2}$ transitions are included).

The pump photon flux is expressed as

$$J_{\mathcal{P}} = \frac{dN_{ph,\mathcal{P}}}{dtdS} = \frac{2\epsilon_0 c |\mathcal{P}^2|}{\hbar\omega_{\mathcal{P}}}. \quad (10)$$

Realisation for Cu-K $_{\alpha 1}$ case

The Eq.(4) can be formulated in an explicit index form (that can be directly implemented by means of `numpy.einsum()` operator). Using Einstein summation convention (for clearness the summation indexes are denoted by $'$), the case of Cu-K $_{\alpha 1}$ (transitions between 6 level) takes the following form

Atomic equations:

$$\frac{\partial}{\partial t} \rho_{ij\mathbf{xyzt}} = \left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{pump/decay}} + \left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{ion.}} + \left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{M.-B.}} + \left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{noise}}, \quad (11)$$

here the index $\mathbf{xyzt}$ corresponds to position in the spacial-temporal grid,

The **pump and decay** part can be represented as:

$$\left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{pump/decay}} = -M_{ij\mathbf{xyzt}}^{(\text{decay})} \rho_{ij\mathbf{xyzt}} + \delta_{ij} \left(p_{i\mathbf{xyzt}}^{(\text{pump})} + \Gamma_{sp} G_{is'}^{(\text{rad.})} \rho_{s's'\mathbf{xyzt}} \right), \quad (12)$$

here the decay part is:

$$M_{ij\mathbf{xyzt}}^{(\text{decay})} = \Gamma_{L3} e_i^{(L3)} e_j^{(L3)} + \Gamma_K e_i^{(K)} e_j^{(K)} + \frac{\Gamma_{L3} + \Gamma_K}{2} \left(e_i^{(L3)} e_j^{(K)} + e_i^{(K)} e_j^{(L3)} \right), \quad (13)$$

the pump part is:

$$p_{i\mathbf{xyzt}}^{(\text{pump})} = \rho_{\mathbf{xyzt}}^{(\text{ground})} J_{\mathcal{F}'\mathbf{xyzt}} S_{\mathcal{F}'i}^{(\text{ground})}, \quad (14)$$

where the sum over $\mathcal{F}'$ runs over $\mathcal{P}, \Omega_{-1}, \Omega_{+1}$. The cross sections for photoionization from the ground state are:

$$S_{\mathcal{F}i}^{(\text{ground})} = \begin{pmatrix} 0.27 \sigma_{\mathcal{P},2p}^{(\text{g})} & 0.23 \sigma_{\mathcal{P},2p}^{(\text{g})} & 0.23 \sigma_{\mathcal{P},2p}^{(\text{g})} & 0.27 \sigma_{\mathcal{P},2p}^{(\text{g})} & 0.5 \sigma_{\mathcal{P},1s}^{(\text{g})} & 0.5 \sigma_{\mathcal{P},1s}^{(\text{g})} & \sigma_{\mathcal{P},o}^{(\text{g})} \\ 0.12 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0.18 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0.28 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0.42 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0 & 0 & \sigma_{\mathcal{E},o}^{(\text{g})} \\ 0.42 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0.28 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0.18 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0.12 \sigma_{\mathcal{E},2p}^{(\text{g})} & 0 & 0 & \sigma_{\mathcal{E},o}^{(\text{g})} \end{pmatrix}, \quad (15)$$

where $\sigma_{\mathcal{F},nl}^{(\text{g})}$ is the total cross section for photoionization from ground state to level nl with field $\mathcal{F}$. Here the sequence of levels is

$$2p_{\frac{3}{2},m=-\frac{3}{2}}, 2p_{\frac{3}{2},m=-\frac{1}{2}}, 2p_{\frac{3}{2},m=\frac{1}{2}}, 2p_{\frac{3}{2},m=\frac{3}{2}}, 1s_{\frac{1}{2},m=-\frac{1}{2}}, 1s_{\frac{1}{2},m=\frac{1}{2}}, (\text{other}), \quad (16)$$

where (other) is an additional level representing the population of states of Cu^+ other than $1s_{\frac{1}{2},m}$ and $2p_{\frac{3}{2},m}$. In all cross-section matrices the sequence of fields is $\mathcal{P}, \Omega_{-1}, \Omega_{+1}$.

The radiative spontaneous transitions:

$$G_{ij}^{(\text{rad.})} = \begin{pmatrix} 0 & 0 & 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{9} & \frac{1}{9} \\ 0 & 0 & 0 & 0 & \frac{1}{9} & \frac{2}{9} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (17)$$

the auxiliary vectors to denote the K and L3 levels:

$$\begin{aligned} e_i^{(L3)} &= (1, 1, 1, 1, 0, 0), \\ e_i^{(K)} &= (0, 0, 0, 0, 1, 1), \end{aligned} \quad (18)$$

here the following sequence of levels is assumed:

$$2p_{\frac{3}{2}, m=-\frac{3}{2}}, 2p_{\frac{3}{2}, m=-\frac{1}{2}}, 2p_{\frac{3}{2}, m=\frac{1}{2}}, 2p_{\frac{3}{2}, m=\frac{3}{2}}, 1s_{\frac{1}{2}, m=-\frac{1}{2}}, 1s_{\frac{1}{2}, m=\frac{1}{2}} \quad (19)$$

The **ionization** part (decay of ionic states due to photoionization) can be represented as:

$$\left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{ion.}} = -\frac{1}{2} \left(\gamma_{i\mathbf{xyzt}}^{(\text{ion.})} + \gamma_{j\mathbf{xyzt}}^{(\text{ion.})} \right) \rho_{ij\mathbf{xyzt}}, \quad (20)$$

$$\gamma_{i\mathbf{xyzt}}^{(\text{ion.})} = S_{\mathcal{F}i}^{(\text{ion.})} J_{\mathcal{F}'\mathbf{xyzt}}, \quad (21)$$

where the cross sections are:

$$S_{\mathcal{F}i}^{(\text{ion.})} = \begin{pmatrix} 1.05 \sigma_{\mathcal{P}, 2p}^{(i)} & 0.95 \sigma_{\mathcal{P}, 2p}^{(i)} & 0.95 \sigma_{\mathcal{P}, 2p}^{(i)} & 1.05 \sigma_{\mathcal{P}, 2p}^{(i)} & \sigma_{\mathcal{P}, 1s}^{(i)} & \sigma_{\mathcal{P}, 1s}^{(i)} \\ 0.70 \sigma_{\mathcal{E}, 2p}^{(i)} & 0.83 \sigma_{\mathcal{E}, 2p}^{(i)} & 1.06 \sigma_{\mathcal{E}, 2p}^{(i)} & 1.41 \sigma_{\mathcal{E}, 2p}^{(i)} & 0.75 \sigma_{\mathcal{E}, 1s}^{(i)} & 1.25 \sigma_{\mathcal{E}, 1s}^{(i)} \\ 1.41 \sigma_{\mathcal{E}, 2p}^{(i)} & 1.06 \sigma_{\mathcal{E}, 2p}^{(i)} & 0.83 \sigma_{\mathcal{E}, 2p}^{(i)} & 0.70 \sigma_{\mathcal{E}, 2p}^{(i)} & 1.25 \sigma_{\mathcal{E}, 1s}^{(i)} & 0.75 \sigma_{\mathcal{E}, 1s}^{(i)} \end{pmatrix}, \quad (22)$$

and $\sigma_{\mathcal{F}, nl}^{(i)}$ is the total cross section for photoionization of level nl with field $\mathcal{F}$.

The **Maxwell-Bloch** part can be represented as:

$$\left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{M.-B.}} = i \left(H_{is'\mathbf{xyzt}}^{\text{int.}} \rho_{s'j\mathbf{xyzt}} - \rho_{is'\mathbf{xyzt}} H_{s'j\mathbf{xyzt}}^{\text{int.}} \right), \quad (23)$$

$$H_{ij\mathbf{xyzt}}^{\text{int.}} = T_{ij\sigma'}^{(+)} \Omega_{\sigma'\mathbf{xyzt}}^{(+)} + T_{ij\sigma'}^{(-)} \Omega_{\sigma'\mathbf{xyzt}}^{(-)}, \quad (24)$$

$$T_{ij\sigma} = \begin{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{3}} \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{3} \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{3} \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{3}} \end{pmatrix} & \begin{pmatrix} \frac{1}{\sqrt{3}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ \frac{1}{3} \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{3}} \\ 0 \\ 0 \end{pmatrix} \end{pmatrix}, \quad (25)$$

$$T_{ij\sigma}^{(+)} = T_{ij\sigma} H_{ij}, \quad T_{ij\sigma}^{(-)} = T_{ij\sigma} H_{ji}, \quad (26)$$

here the sequence of polarizations is $-1, +1$.

The **noise** part can be represented as:

$$\left(\frac{\partial \rho_{ij\mathbf{xyzt}}}{\partial t} \right)_{\text{noise}} = \sqrt{\frac{\gamma_{\text{decoh.}}}{T_{i'j'\sigma}^{(-)} \rho_{j'k'\mathbf{xyzt}} T_{k'i'\sigma}^{(+)} n \Delta x \Delta y \Delta z \Delta t}} \left(T_{is'\sigma'}^{(-)} \xi_{\sigma'\mathbf{xyzt}}^{(+)*} \rho_{s'j\mathbf{xyzt}} + \rho_{is'\mathbf{xyzt}} T_{s'j\sigma'}^{(+)} \xi_{\sigma'\mathbf{xyzt}}^{(-)*} \right), \quad (27)$$

$$\gamma_{\text{decoh.}} = \frac{\Gamma_{L3} + \Gamma_K}{2}.$$

The population of the atomic ground state evolves according to:

$$\frac{\partial}{\partial t} \rho_{\mathbf{xyzt}}^{(\text{ground})} = -\rho_{\mathbf{xyzt}}^{(\text{ground})} S_{\mathcal{F}'i'}^{(\text{ground})} J_{\mathcal{F}'\mathbf{xyzt}} \quad (28)$$

The population of the additional level (**other**) is described by

$$\frac{\partial}{\partial t} \rho_{\mathbf{xyzt}}^{(\text{other})} = -\rho_{\mathbf{xyzt}}^{(\text{other})} S_{\mathcal{F}'}^{(\text{other})} J_{\mathcal{F}'\mathbf{xyzt}} + \rho_{\mathbf{xyzt}}^{(\text{ground})} S_{\mathcal{F}'7}^{(\text{ground})} J_{\mathcal{F}'\mathbf{xyzt}} \quad (29)$$

where the cross sections for photoionization of level (other) are:

$$S_{\mathcal{F}}^{(\text{other})} = \begin{pmatrix} \sigma_{\mathcal{P}}^{(\text{o})} \\ \sigma_{\mathcal{E}}^{(\text{o})} \\ \sigma_{\mathcal{E}}^{(\text{o})} \end{pmatrix} \quad (30)$$

Field equations

Using the same conventions, the field equations are:

$$\left[\frac{\partial}{\partial z} - \frac{i}{2k_0} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \Omega_{\sigma\text{xyzt}}^{(+)} = -\frac{\kappa_{\Omega_{\sigma\text{xyzt}}}}{2} \Omega_{\sigma\text{xyzt}}^{(+)} + i \frac{3}{8\pi} \lambda^2 \Gamma_{sp} \left(n T_{i'j'\sigma}^{(-)} \frac{\rho_{j'i'\text{xyzt}} + \rho_{i'j'\text{xyzt}}^*}{2} + \sqrt{\frac{T_{i'j'\sigma}^{(-)} \rho_{j'k'\text{xyzt}} T_{k'i'\sigma}^{(+)}}{\gamma_{\text{decoh.}} \Delta x \Delta y \Delta z \Delta t}} n \xi_{\sigma\text{xyzt}}^{(+)} \right) \quad (31)$$

$$\left[\frac{\partial}{\partial z} + \frac{i}{2k_0} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \Omega_{\sigma\text{xyzt}}^{(-)} = -\frac{\kappa_{\Omega_{\sigma\text{xyzt}}}}{2} \Omega_{\sigma\text{xyzt}}^{(-)} - i \frac{3}{8\pi} \lambda^2 \Gamma_{sp} \left(n T_{j'i'\sigma}^{(+)} \frac{\rho_{i'j'\text{xyzt}} + \rho_{j'i'\text{xyzt}}^*}{2} + \sqrt{\frac{T_{i'j'\sigma}^{(-)} \rho_{j'k'\text{xyzt}} T_{k'i'\sigma}^{(+)}}{\gamma_{\text{decoh.}} \Delta x \Delta y \Delta z \Delta t}} n \xi_{\sigma\text{xyzt}}^{(-)} \right) \quad (32)$$

$$\left[\frac{\partial}{\partial z} - \frac{i}{2k_{\mathcal{P}}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \right] \mathcal{P}_{\text{xyzt}} = -\frac{\kappa_{\mathcal{P}_{\text{xyzt}}}}{2} \mathcal{P}_{\text{xyzt}} \quad (33)$$

The absorption coefficients can be represented as:

$$\kappa_{\mathcal{F}} = n \left(\rho_{\text{xyzt}}^{(\text{ground})} S_{\mathcal{F}i'}^{(\text{ground})} + \rho_{\text{xyzt}}^{(\text{other})} S_{\mathcal{F}}^{(\text{other})} + \rho_{i'i'\text{xyzt}} S_{\mathcal{F}i'}^{(\text{ion.})} + \sigma_{\mathcal{F}}^{(\text{compound})} \right), \quad (34)$$

$$\sigma_{\mathcal{F}}^{(\text{compound})} = \sum_{\text{el.}} N_{\text{el.}} \sigma_{\mathcal{F},\text{el.}} \quad (35)$$

is the absorption due to the elements of the compound other than copper, where $\sum_{\text{el.}}$ runs over these elements, $N_{\text{el.}}$ denotes the number of atoms of given element per one copper atom, and $\sigma_{\mathcal{F},\text{el.}}$ is the corresponding cross section for photoionization with field $\mathcal{F}$.

Some relations to cross-check the limiting case of spontaneous emission

The amount of photons emitted due to spontaneous fluorescence:

$$\frac{dN_{ph.,sp.}}{dt dS} = \frac{3}{8\pi} \int d^3 \vec{r}' n \lambda^2 \Gamma_{sp} T_{is\sigma}^{(-)} \rho_{sj}(\vec{r}', t) T_{ji\sigma}^{(+)} |G(|\vec{r} - \vec{r}'|)|^2, \quad (36)$$

here $G(|\vec{r} - \vec{r}'|)$ is Green's function for propagation of the emitted field:

$$G(R) = \frac{z}{i\lambda} \frac{e^{ik_0 R}}{R^2} \simeq \frac{e^{\frac{ik_0(x^2+y^2)}{2z}}}{i\lambda z}, \quad (37)$$

here in the last relation the paraxial approximation was used.